Walid Hemati

ML & Software Engineer · Computer vision, Android · Freiburg

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PROFILE

ML and software engineer. I work on computer vision for broadcast and streamed video at Shikenso in Freiburg, build Android apps that keep data on the device, and write scientific code in Python and C++. B.Sc. in Computer Engineering from Istanbul Kültür University; studying at the University of Freiburg.

EXPERIENCE

Current
Shikenso, Freiburg
Computer vision on broadcast and streamed video.

07/2021 – 09/2021
Web developer intern, Miss Diamond, Istanbul
Developed the company website; maintained its computers and website.

EDUCATION

Current
University of Freiburg, Freiburg
Student

07/2019 – 07/2023
Istanbul Kültür University, Istanbul
B.Sc. Computer Engineering

2017 – 2019
Beşir Balcıoğlu Anadolu Lisesi, Istanbul
High school

2008 – 2017
Sharjah Indian School, Sharjah, UAE
School

PROJECTS

Open source
pylbm · Scientific computing
A GPU Lattice-Boltzmann solver in PyTorch/CUDA for the D2Q9 lid-driven cavity, validated against Ghia et al. (1982).
Python · PyTorch · CUDA

In development
SteuerApp · Android · Privacy-first
An offline German tax recorder for Android. Receipts are read and classified on the device; by default nothing leaves the phone.
Kotlin · Android · On-device OCR · Room + SQLCipher

Android
Flatsnap · Android · On-device OCR
An on-device document scanner for Android. It finds the page edges, corrects the perspective and reads the text with ML Kit, all on the phone.
Android · ML Kit

Research
Emotion from EEG · Research · ML
Biosignal-based latent factor decoding for emotion detection with a variational autoencoder, on the DEAP and SEED datasets.
Python · MATLAB

Open source

Turkish review classification · NLP

Classifying Turkish Hepsiburada reviews, comparing logistic regression and SVM with BERTurk and DistilBERT.

Python · Jupyter · BERT

Coursework

yamd · Scientific computing · C++

Coursework for the University of Freiburg course Molecular Dynamics in C++, built on the course's Meson template with Eigen and GoogleTest.

C++ · Eigen · gtest · Meson

Review paper

Machine Learning Based Android Malware Detection: A State of Art

A review of machine-learning approaches to detecting malware on Android.

SKILLS

Daily — Python · PyTorch · computer vision on video · Git

Comfortable — Kotlin / Android · C++ (Eigen, Meson, gtest) · CUDA via PyTorch · MATLAB · SQL

Have used — C · Java · C# · Flutter / Dart · JavaScript · Solidity / Truffle

LANGUAGES

Dari (native) · English (C1) · Persian (B2 listening, B1 spoken) · Turkish (B1) · Urdu / Hindi (B1 spoken) · Arabic (A1–A2)